Owais Ali Khan

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Karachi, Pakistan | Open to relocation (GCC) & remote

SKILLS

LANGUAGES
Java • Python • C/C++ • SQL • Go • Bash • Groovy

BACKEND
REST APIs • Django / DRF • FastAPI • gunicorn • Microservices • Multi-tenant systems

DATA & MESSAGING
PostgreSQL • MySQL • Redis • Kafka • Elasticsearch

DEVOPS & CLOUD
Docker • Kubernetes • Jenkins (CI/CD) • AWS (S3, ECS, EKS, ECR) • NGINX • Git • Linux

RELIABILITY & SECURITY
OOM & crash-loop RCA • Load testing (Locust) • CVE remediation (Trivy, Grype) • Container hardening

EDUCATION

FAST NUCES, KARACHI
• Bachelors of Science in Computer Science, CGPA: 3.88

ACHIEVEMENTS

- Gold Medallist ICPC Asia Topi Regional '24-25 (National 4th rank)
- Silver Medallist ICPC Asia Topi Regional '23-24 (National 8th rank)
- IEEE Xtreme 17.0 (Global Rank 531/4200)
- Rector's List (Spring '22)
- Dean's List (7 semesters, Fall '21 -- Spring '25)

VOLUNTEERING & COMMUNITIES

- Student Lab Assistant, DSA (Fall '24, Fall '25)
- ACM Devday'25 Mentor (Jan '25 - Apr '25)
- Procom'24 Co-head CS Competitions (Jan '24 - Mar '24)

EXPERIENCE

SECURITI AI | Associate Backend Engineer
June '25 - Present | Hybrid

- Build and operate backend **Java** and **Python** microservices that scan files and *multi-tenant* **PostgreSQL**/**MySQL** databases for sensitive data, serving enterprise customers under strict *SLAs*.
- Shipped a new **gunicorn**-based **Python** service end-to-end — multi-stage **Docker** builds, **Jenkins** CI/CD and **Kubernetes** deployment — increasing pipeline throughput and horizontal scalability.
- Re-architected ephemeral data storage on **Redis** (hashes over serialized JSON), cutting payload size and retry churn on the hot path.
- Led the **JDK 11** → **17** migration and delivered the **Python 3.9** → **3.11** upgrade across both services, owning the dependency fallout and release-over-release regression validation.
- Root-caused production **OOMs** via streaming extraction and memory-footprint reduction, and eliminated pod *CrashLoopBackOffs* under scale testing through **gunicorn** worker-lifecycle fixes.
- Hardened **Kubernetes** workloads (non-root, read-only FS, dropped capabilities) and drove ongoing **CVE remediation** across all service images.

SECURITI AI | Software Engineer Intern
June '24 - Aug '24 | Hybrid

- Engineered high-performance infrastructure for large-scale *regular-expression* matching using **Intel Hyperscan**, achieving speedups of 2–3 orders of magnitude.

RETROCAUSAL AI | Applied Research Intern
Mar '24 - June '24 | Remote

- Developed *RESTful* API endpoints and automated reporting workflows using the **Django REST Framework**, with scalable storage and retrieval on **Amazon S3**.

LDP LOGISTICS | IT Support Specialist
June '23 - Sep '23 | Onsite

- Built and deployed a full-stack *web application* (**Django**, **Bootstrap4**) for managing electronic *inventory* across local and overseas offices.

PROJECTS

LDP ASSET MANAGER | Fullstack Web Application
Summer 2023 | Full Stack Development

- An inventory-management *web portal* (**Django**, **MySQL**, **MUI**) tracking assets, accessories and licenses across all LDP offices.

LLM TEXT DETECTOR | Final Year Project
Fall 2024, Spring 2025 | Natural Language Processing

- **A Hybrid Approach to LLM-generated Text Detection**: combined *cross-perplexity* scores with a fine-tuned *T5* transformer (**TensorFlow**/**Keras**), reaching a 94% **F1-score** on a curated dataset.